

A Proof of the Tratnik–Ye Medianity Conjecture for Resonance Graphs of Perfect Matchings on Surfaces*

GUANGFU WANG AND KE KANG

School of Mathematics and Information Sciences, Yantai University
Yantai, Shandong Province, 264005, China
E-mail address: gfwang@ytu.edu.cn

Abstract. Tratnik and Ye conjectured that if G is a graph embedded in a closed surface and $\mathcal{F} \subsetneq F(G)$ is a set of even faces, then every connected component of the resonance graph $R(G; \mathcal{F})$ is a median graph. We prove the conjecture. For each connected component H we construct a cubical complex X_H whose 1-skeleton, namely the graph formed by its vertices and edges, is H ; its cubes are generated by pairwise boundary-disjoint facial rotations. The vertex links of X_H are flag. It remains to prove simple connectedness. We do this by reducing closed words in facial rotations. The main local ingredient is a parity-propagation lemma: if an even face f is alternating before and after a word of facial rotations in which f does not occur, then the occurrence parities of all face sectors incident with ∂f , except the sector of f , are forced to be equal. A shortest non-null-homotopic loop would then produce an infinite strictly descending chain of nonempty subwords, which is impossible. Hence X_H is a CAT(0) cube complex and $H = X_H^{(1)}$ is median.

2020 Mathematics Subject Classification. 05C70, 05C10, 05C62, 57M15.

Key words and phrases. Perfect matching; resonance graph; median graph; CAT(0) cube complex; graph on a surface.

1 Introduction

Let G be a finite graph embedded in a connected closed surface Σ , and let $F(G)$ denote the set of faces of the embedding. We use the standard strong-embedding conventions used by Tratnik and Ye [6]: every face considered below is bounded by a facial cycle, and the face-dual used in parity arguments is connected. Thus, if an edge e has the two incident face-sides belonging to faces a and b , then toggling facial boundaries gives the usual mod-2 relation between the occurrence parities of a and b . In particular, for a facial cycle ∂f and an edge of ∂f , there is a well-defined face on the other side of that boundary edge.

A face f is *even* if its boundary ∂f is an even cycle. For a set $\mathcal{F} \subseteq F(G)$ of even faces, the *resonance graph* $R(G; \mathcal{F})$ has as vertices the perfect matchings of G ; two perfect matchings M and N are adjacent if and only if

$$M \triangle N = E(f)$$

for some $f \in \mathcal{F}$. Equivalently, N is obtained from M by rotating, or flipping, the M -alternating facial cycle ∂f . Concretely, this means that along ∂f the matching M uses every other boundary edge; the flip removes those used boundary edges and inserts the complementary boundary edges of ∂f , leaving all edges outside ∂f unchanged. The result is again a perfect matching because ∂f is an even alternating cycle.

*This work was supported by the Natural Science Foundation of Shandong Province (Grant No. ZR2024MA073).

Tratnik and Ye proved that, when $\mathcal{F} \subsetneq F(G)$, every connected component of $R(G; \mathcal{F})$ is an induced subgraph of a hypercube [6, Theorem 3.5]. They also observed that their embedding need not be isometric and formulated the following conjecture [6, Conjecture 5.3].

Theorem 1.1 (Tratnik–Ye medianity conjecture). *Let G be a graph embedded in a closed surface, and let $\mathcal{F} \subsetneq F(G)$ be a set of even faces. Then every connected component of $R(G; \mathcal{F})$ is a median graph.*

Recall that a connected graph is *median* if, for every three vertices x, y, z , there is a unique vertex lying simultaneously on a shortest x – y path, a shortest y – z path, and a shortest z – x path. Theorem 1.1 extends the familiar medianity of resonance graphs in several planar and chemical-graph settings, where the conclusion often follows from a distributive-lattice structure; see, for example, [5] and the discussion in [6, 7]. The point here is that no lattice structure is assumed or constructed. Instead, we attach cubes to independent facial rotations and use Gromov’s criterion for CAT(0) cube complexes.

A cube complex is a space obtained by gluing Euclidean cubes along their faces. We shall use only its combinatorial data: vertices, edges, square faces, higher-dimensional cubes, and the local links at vertices.

We shall use two standard facts. First, Gromov’s link criterion says that a cube complex is CAT(0) if and only if it is simply connected and all vertex links are flag simplicial complexes; see [3, Chapter II.5]. Second, the 1-skeleton of a CAT(0) cube complex is a median graph, as in Chepoi’s characterization [4].

Notation and terminology

For reference, we spell out the notation used throughout the proof.

Σ A connected closed surface in which the finite graph G is embedded.

$V(G), E(G), F(G)$

The vertex set, edge set, and face set of the embedded graph G .

$\partial f, V(\partial f), E(f)$

For a face $f \in F(G)$, ∂f is its facial boundary cycle, $V(\partial f)$ is the set of vertices on this cycle, and $E(f)$ is the set of boundary edges of this cycle.

$\mathcal{F}$ A fixed proper subset of $F(G)$ consisting of even faces. The condition $\mathcal{F} \subsetneq F(G)$ is used essentially in Lemma 2.1 and Theorem 5.2.

M, N

Perfect matchings of G , viewed as subsets of $E(G)$. A face f is *M -alternating* if, along the cycle ∂f , the edges alternate between belonging to M and not belonging to M .

$\triangle$ and $\triangle$

Symmetric difference of sets. Thus $M \triangle N$ is the set of edges belonging to exactly one of M, N , and $M \triangle E(f)$ is obtained from M by toggling the boundary edges of f . The symbol $\triangle_{f \in T} E(f)$ denotes the iterated symmetric difference of the edge sets $E(f)$ over $f \in T$.

$R(G; \mathcal{F})$

The resonance graph whose vertices are the perfect matchings of G and whose edges are the allowed facial rotations along faces in $\mathcal{F}$.

face-dual

The graph whose vertices are the faces in $F(G)$ and whose edges join two faces sharing a boundary edge. We use connectedness of this graph in the parity arguments.

face sector

At a vertex of the embedded graph, a face sector is the sector between two consecutive incident half-edges in the cyclic order determined by the embedding.

$M \xrightarrow{f} N$

A labeled step in $R(G; \mathcal{F})$: it means that $f \in \mathcal{F}$ is M -alternating and $N = M \triangle E(f)$. The arrow records the direction in which a path is read and the face label of the step; it does not impose an orientation on the resonance graph.

H A connected component of $R(G; \mathcal{F})$.

X_H and $X_H^{(1)}$

The cubical complex built from the component H in Section 3, and its 1-skeleton. For any cube complex X , the 1-skeleton $X^{(1)}$ is the ordinary graph obtained by keeping all vertices and edges of X and forgetting all cubes of dimension at least 2. We will have $X_H^{(1)} = H$.

$\text{Lk}_X(x)$

The link of a vertex x in a cube complex X . Its vertices are germs of edges of X issuing from x ; an edge germ is the local direction of an edge starting at x . A set of such vertices spans a simplex exactly when those edge germs lie in a common cube of X . In particular, $\text{Lk}_{X_H}(M)$ denotes the link of the vertex M in X_H .

$\varepsilon_u(h)$

For a word u in face labels and a face h , this is the parity, in $\{0, 1\}$, of the number of occurrences of h in u .

$\text{supp}(w)$

The support of a face word w , namely the set of distinct face labels that occur in w .

clique, simplex, flag

A clique in a graph is a set of vertices that are pairwise adjacent. A k -simplex is the combinatorial analogue of a filled k -dimensional triangle; for example, a 1-simplex is an edge and a 2-simplex is a filled triangle. A simplicial complex is flag if every finite clique in its 1-skeleton is filled in as a simplex.

Two faces f, g are called *disjoint* if their boundary cycles are vertex-disjoint, i.e. $\partial f \cap \partial g = \emptyset$. They *meet* if their boundary cycles have a common vertex. If a resonance edge admits more than one facial representative with the same boundary-edge set, we fix one representative label once and for all whenever we write a face word. This convention affects only notation, not the rotation itself.

2 Closed rotation words and parity

Let

$$W = M_0 M_1 \cdots M_t$$

be a path in $R(G; \mathcal{F})$, meaning that consecutive matchings M_{i-1} and M_i are adjacent in the resonance graph. A *face word* for W is a sequence

$$w = f_1 f_2 \cdots f_t, \quad f_i \in \mathcal{F},$$

where juxtaposition denotes a word, not multiplication, such that

$$M_{i-1} \triangle M_i = E(f_i), \quad i = 1, \dots, t.$$

If W is closed, i.e. if $M_t = M_0$, then w is called a *closed face word*. For a word u and a face h , let

$$\varepsilon_u(h) \in \{0, 1\}$$

denote the parity of the number of occurrences of h in u .

The following is the closed-walk form of the parity lemma of Tratnik and Ye. We include the proof because we shall apply it to closed walks with repetitions, not only to simple cycles.

Lemma 2.1. *Let $\mathcal{F} \subsetneq F(G)$, and let $w = f_1 \cdots f_t$ be a closed face word in $R(G; \mathcal{F})$. Then every face of G occurs in w an even number of times. In particular, every face of $\mathcal{F}$ that occurs in w occurs at least twice.*

Proof. Let the corresponding closed path be

$$M_0 M_1 \cdots M_t = M_0.$$

Since $M_{i-1} \triangle M_i = E(f_i)$ for each i , we have

$$E(f_1) \triangle E(f_2) \triangle \cdots \triangle E(f_t) = \emptyset.$$

Hence, for each edge e of G , the number of selected facial boundaries containing e is even. If the two face-sides incident with e belong to faces a and b , this gives

$$\varepsilon_w(a) + \varepsilon_w(b) \equiv 0 \pmod{2},$$

and therefore $\varepsilon_w(a) = \varepsilon_w(b)$. By connectedness of the face-dual, all faces of G have the same occurrence parity. Since $\mathcal{F}$ is a proper subset of $F(G)$, some face $q \notin \mathcal{F}$ never occurs in w ; thus $\varepsilon_w(q) = 0$. Therefore every face has parity 0. $\square$

The notation $M \xrightarrow{f} N$ used below means that $N = M \triangle E(f)$ and that ∂f is alternating with respect to the matching at the tail of the arrow. Thus a displayed chain of arrows is a path in the resonance graph, with the labels recording the successive faces flipped.

Lemma 2.2. *Let $f, g \in \mathcal{F}$ be disjoint faces. If*

$$M \xrightarrow{f} M \triangle E(f) \xrightarrow{g} M \triangle E(f) \triangle E(g)$$

is a path in $R(G; \mathcal{F})$, then

$$M \xrightarrow{g} M \triangle E(g) \xrightarrow{f} M \triangle E(g) \triangle E(f)$$

is also a path. Thus the two length-two paths bound a square.

Proof. Flipping f changes only the membership of edges of ∂f , and flipping g changes only the membership of edges of ∂g . Since ∂f and ∂g are vertex-disjoint, flipping one of them does not affect the alternating status of the other. Hence the two rotations commute. $\square$

3 The cubical complex associated with a component

Let H be a connected component of $R(G; \mathcal{F})$. We construct a cube complex X_H as follows. Its vertices are the vertices of H , i.e. the perfect matchings in this component, and its edges are the edges of H .

More generally, suppose that $M \in V(H)$ and that

$$S = \{f_1, \dots, f_k\} \subseteq \mathcal{F}$$

is a set of pairwise disjoint faces such that every f_i is M -alternating. The rotations in S can be performed independently. For each $T \subseteq S$, put

$$M_T = M \Delta (\Delta_{f \in T} E(f)).$$

The 2^k matchings M_T , $T \subseteq S$, span a Boolean k -cube in H : the subsets $T \subseteq S$ play the role of the binary vectors in $\{0, 1\}^k$, and two vertices M_T and $M_{T'}$ are adjacent exactly when T and T' differ by one face. Here M_T is the matching obtained from M by flipping exactly the faces in T . We attach one Euclidean k -cube for each such Boolean family of vertices, identifying duplicate descriptions of the same cube. Equivalently, X_H is the cubical complex whose cubes are precisely the Boolean subgraphs obtained from simultaneously flippable, pairwise disjoint facial cycles.

The 1-skeleton $X_H^{(1)}$ is the ordinary graph obtained from X_H by keeping only its vertices and one-dimensional cubes. Thus attaching a square, a three-dimensional cube, or a higher cube records extra homotopies among paths, but it does not create new vertices or new graph edges. In the present construction, the vertices and edges are exactly those of the component H .

For a cube complex X and a vertex $x \in V(X)$, the notation

$$\text{Lk}_X(x)$$

denotes the *link* of x in X . This is a simplicial complex recording the local directions at x . Its vertices are the germs of edges of X issuing from x ; here a germ may be read simply as the direction of an edge as it leaves x . Several such link vertices span a simplex precisely when the corresponding directions are contained in one common cube with initial vertex x .

For example, if X is a filled square and x is a corner, then two edges leave x and they lie in the same square; hence $\text{Lk}_X(x)$ consists of two vertices joined by an edge. If the same two edges form only an “L” shape and the square is not filled in, then the link still has two vertices, but they are not joined. If X is a solid three-dimensional cube and x is a corner, then the three coordinate directions from x span a filled triangle in $\text{Lk}_X(x)$, because all three directions lie in one common 3-cube.

In our complex X_H , a vertex is a perfect matching M . An edge germ at M is a possible first flip

$$M \xrightarrow{f} M \Delta E(f).$$

Thus the vertices of $\text{Lk}_{X_H}(M)$ are the facial rotations that are legal at M . A collection of such link vertices spans a simplex exactly when the corresponding facial rotations can be performed simultaneously, equivalently when they generate one cube of X_H . A simplicial complex is *flag* if every finite clique in its 1-skeleton spans a simplex; in other words, pairwise compatibility of directions already implies simultaneous compatibility.

Lemma 3.1. *The 1-skeleton of X_H is H , and every vertex link of X_H is a flag simplicial complex.*

Proof. The 1-skeleton is H by construction. Fix a vertex M of X_H . By the definition of the link, the vertices of $\text{Lk}_{X_H}(M)$ are precisely the edge germs at M in X_H . These edge germs are represented by the facial rotations $f \in \mathcal{F}$ that are M -alternating and whose flip gives an edge of H ; here we use the fixed representative labels chosen in the introduction. A set of such link vertices spans a simplex exactly when the corresponding faces are pairwise disjoint. Indeed, if they are pairwise disjoint, Lemma 2.2 shows that their rotations are independent and therefore generate a cube of X_H . Conversely, every cube of X_H was attached from a pairwise disjoint set of facial rotations. Hence a clique in the link means that the rotations are pairwise disjoint; the preceding construction then fills in the whole simplex. Thus every clique in the link spans a simplex, and the link is flag. $\square$

Proposition 3.2. *If X_H is simply connected, then H is a median graph.*

Proof. By Lemma 3.1, X_H satisfies the flag-link condition. If X_H is simply connected, Gromov's criterion implies that X_H is a CAT(0) cube complex [3, Chapter II.5]. The 1-skeleton of a CAT(0) cube complex is a median graph [4]. Since $X_H^{(1)} = H$, the graph H is median. $\square$

It remains to prove that X_H is simply connected.

4 A local parity-propagation lemma

Let f be an even face and let

$$P = N_0 N_1 \cdots N_s$$

be a path in $R(G; \mathcal{F})$ with face word $u = g_1 \cdots g_s$. Assume that f does not occur in u . For a face h , $\varepsilon_u(h)$ denotes the parity of the number of occurrences of h in u ; if h does not occur, this parity is 0.

Lemma 4.1. *Assume that f is N_0 -alternating and N_s -alternating. Let v be a vertex of ∂f . Then all face sectors at v different from the sector of f have the same occurrence parity in u . Equivalently, all faces of G distinct from f that are incident with v have the same parity in u .*

Proof. List the edges incident with v in cyclic order around v as

$$e_0, e_1, \dots, e_{d-1},$$

where e_0 and e_{d-1} are the two boundary edges of f , and where the sector between e_{d-1} and e_0 is the sector of f . For $i = 1, \dots, d-1$, let h_i be the face sector between e_{i-1} and e_i . These are precisely the face sectors at v different from the sector of f . Put

$$\eta_i = \varepsilon_u(h_i),$$

where this is the parity of the face containing the sector h_i . If the same face occurs in more than one sector, the corresponding η_i are the same number.

Each time a face is flipped, it toggles exactly the boundary edges of that face. Therefore, for an edge incident with v , its membership in the matching changes once for each occurrence of one of the two face sectors adjacent to that edge. Counting modulo 2, the parity with which the membership of an incident edge changes from N_0 to N_s is the mod-2 sum of the occurrence parities of the two adjacent face sectors. Since f does not occur in u , the change parities at v are

$$e_0 : \eta_1, \quad e_i : \eta_i + \eta_{i+1} \pmod{2} \quad (1 \leq i \leq d-2), \quad e_{d-1} : \eta_{d-1}.$$

Because f is alternating at v with respect to both N_0 and N_s , exactly one of the two f -boundary edges e_0, e_{d-1} is the matching edge incident with v in each endpoint matching. Hence no edge e_i with $1 \leq i \leq d-2$ changes membership, while the two edges e_0 and e_{d-1} either both change membership or both do not. Therefore

$$\eta_i = \eta_{i+1} \pmod{2} \quad (1 \leq i \leq d-2), \quad \eta_1 = \eta_{d-1}.$$

Thus all η_i are equal, as required. $\square$

Corollary 4.2. *With the hypotheses of Lemma 4.1, the parities $\varepsilon_u(h)$ are equal for all faces $h \neq f$ with $\partial h \cap \partial f \neq \emptyset$. Consequently, if one such face has parity zero in u , then every face meeting f has parity zero in u .*

Proof. By Lemma 4.1, for each vertex $v \in V(\partial f)$ there is a local parity c_v such that every face incident with v and distinct from f has parity c_v . If vv' is an edge of ∂f and r is the face on the other side of that boundary edge, then $r \neq f$ and r is incident with both v and v' . This common neighboring face transmits the equality from the local picture at v to the local picture at v' , giving

$$c_v = \varepsilon_u(r) = c_{v'}.$$

Since the boundary cycle ∂f is connected, all c_v are equal. The conclusion follows. $\square$

5 Reduced closed words

We now translate homotopies in the 2-skeleton of X_H into operations on face words. The 2-skeleton is the subcomplex consisting of all vertices, all edges, and all square 2-faces of X_H . Higher-dimensional cubes do not add new fundamental-group relations beyond those already visible in their square faces, so, by cellular approximation, simple connectedness of X_H is equivalent to null-homotopy of all closed edge paths in $X_H^{(1)} = H$.

There are two elementary moves on face words:

- (i) deletion of a consecutive pair ff . This means that the path flips f and then immediately flips f back, so the two steps are an immediate backtrack along the same resonance edge;
- (ii) interchange of consecutive letters $fg \leftrightarrow gf$ when $\partial f \cap \partial g = \emptyset$. This is the square homotopy from Lemma 2.2: the two disjoint flips can be performed in either order and lead to the same matching.

These moves preserve the homotopy class of the corresponding loop in X_H .

A nonempty closed face word is read cyclically, because a closed loop has no distinguished starting edge. Thus a cyclic conjugate of $f_1 f_2 \cdots f_m$ is obtained by choosing a different starting point, for instance $f_i f_{i+1} \cdots f_m f_1 \cdots f_{i-1}$. A closed face word is called *fully reduced* if no cyclic conjugate of any word obtained from it by a sequence of disjoint commutations contains a consecutive pair ff . A shortest non-null-homotopic edge loop in X_H is necessarily fully reduced: otherwise square homotopies followed by deletion of a backtrack would produce a shorter loop in the same nontrivial homotopy class.

Lemma 5.1. *Let w be a fully reduced closed face word. Suppose that, cyclically,*

$$w = f u f u'$$

where u' is another, possibly empty, word and u contains no occurrence of f . Then some face occurring in u meets f .

Proof. If every face occurring in u were disjoint from f , then repeated use of Lemma 2.2 would commute the first displayed occurrence of f across all letters of u , producing a cyclic word with a consecutive pair ff . This contradicts full reducedness. $\square$

Theorem 5.2. *For every connected component H of $R(G; \mathcal{F})$, the cubical complex X_H is simply connected.*

Proof. Suppose, for a contradiction, that some closed edge path in $H = X_H^{(1)}$ is not null-homotopic in X_H . Choose such a path of minimum length, and let

$$w = f_1 f_2 \cdots f_m$$

be a face word for it. As observed above, w is fully reduced. Put

$$A = \{f_i : 1 \leq i \leq m\}.$$

By Lemma 2.1, each face in A occurs an even number of times. Moreover $A \subseteq \mathcal{F} \subsetneq F(G)$, so A is a proper subset of the full face set. Since the face-dual is connected and A is nonempty, there are faces $p_0 \in A$ and $p_{-1} \notin A$ such that p_{-1} shares a boundary edge with p_0 .

Choose two consecutive cyclic occurrences of p_0 in w and write the corresponding cyclic subword as

$$p_0 U_0 p_0,$$

where U_0 contains no occurrence of p_0 . The face p_0 is alternating at both ends of the path segment represented by U_0 : immediately after a flip of p_0 it is still alternating, with the complementary set of boundary edges selected, and immediately before the second displayed occurrence it is alternating because that second flip is legal. The face p_{-1} does not occur in U_0 , since $p_{-1} \notin A$.

We shall construct inductively faces

$$p_0, p_1, p_2, \dots$$

and nonempty words

$$U_0 \supsetneq U_1 \supsetneq U_2 \supsetneq \cdots,$$

where the symbol $\supsetneq$ means proper containment as contiguous subwords: each U_{i+1} is a proper contiguous subword of U_i . Suppose that p_i and U_i have been constructed with the following properties:

- (a) p_{i-1} meets p_i ;
- (b) p_{i-1} does not occur in U_i ;
- (c) $p_i U_i p_i$ is formed by two consecutive occurrences of p_i inside the previous word (inside the cyclic word w when $i = 0$, and inside U_{i-1} when $i \geq 1$); in particular p_i does not occur in U_i ;
- (d) p_i is alternating at both ends of the path segment represented by U_i .

The initial pair (p_0, U_0) satisfies these properties, with p_{-1} as above.

Apply Corollary 4.2 to the face p_i and the word U_i . Since p_{i-1} meets p_i and has parity zero in U_i , every face meeting p_i has parity zero in U_i . By Lemma 5.1, the word U_i contains at least one face meeting p_i ; choose one and call it p_{i+1} . Its occurrence parity in U_i is zero and it occurs at least once, so it occurs at least twice. Choose two consecutive occurrences of p_{i+1} inside U_i and write the corresponding subword as

$$p_{i+1} U_{i+1} p_{i+1}.$$

Then U_{i+1} contains no occurrence of p_{i+1} and is a proper contiguous subword of U_i . It cannot be empty, for otherwise the original fully reduced cyclic word w would contain a consecutive pair $p_{i+1} p_{i+1}$. Finally, p_i does not occur in U_{i+1} because U_i contains no occurrence of p_i . The face p_{i+1} is alternating at both ends of the path segment represented by U_{i+1} : it is alternating immediately after the first displayed flip of p_{i+1} , and it is alternating immediately before the second displayed flip because that flip is legal.

Thus the induction continues forever and produces an infinite strictly descending chain of nonempty finite words. This is impossible because their lengths strictly decrease. Hence no non-null-homotopic closed edge path exists, and X_H is simply connected. $\square$

6 Proof of the main theorem and final remarks

Proof of Theorem 1.1. Let H be a connected component of $R(G; \mathcal{F})$. If H consists of a single vertex, then it is a median graph. Otherwise construct X_H as in Section 3. By Lemma 3.1, all vertex links of X_H are flag. By Theorem 5.2, X_H is simply connected. Therefore X_H is a CAT(0) cube complex by Gromov's criterion, and its 1-skeleton is a median graph. Since $X_H^{(1)} = H$, the component H is median. $\square$

Remark 6.1. *The assumption $\mathcal{F} \subsetneq F(G)$ is used in two essential places. First, it gives the parity-of-labels statement for closed face words, Lemma 2.1. Second, for the support A of a putative minimal non-null-homotopic loop, it guarantees a face outside A ; connectedness of the face-dual then supplies the initial adjacent pair $p_{-1} \notin A$ and $p_0 \in A$ that starts the descending-word argument. This also explains why the case $\mathcal{F} = F(G)$ is genuinely different.*

Remark 6.2. *The proof does not use the particular induced-hypercube embedding constructed in [6]. The cubical complex X_H is determined intrinsically by independent facial rotations. Medianity follows from nonpositive curvature of this complex, not from an a priori isometric embedding of H into a hypercube.*

References

- [1] S. P. Avann, *Metric ternary distributive semi-lattices*, Proc. Amer. Math. Soc. **12** (1961), 407–414.
- [2] H.-J. Bandelt, *Retracts of hypercubes*, J. Graph Theory **8** (1984), 501–510.
- [3] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Grundlehren der Mathematischen Wissenschaften, vol. 319, Springer, Berlin, 1999.
- [4] V. Chepoi, *Graphs of some $CAT(0)$ complexes*, Adv. in Appl. Math. **24** (2000), 125–179.
- [5] P. C. B. Lam and H. Zhang, *A distributive lattice on the set of perfect matchings of a plane bipartite graph*, Order **20** (2003), 13–29.
- [6] N. Tratnik and D. Ye, *Resonance graphs on perfect matchings of graphs on surfaces*, Graphs Combin. **39** (2023), Paper No. 68, 15 pp.
- [7] N. Tratnik and P. Žigert Pleteršek, *Resonance graphs of fullerenes*, Ars Math. Contemp. **11** (2016), 425–435.